```
%p, q ,a constants
a = 0.75;
p = 1.5;
q = 1.25;

%bvalues given in question
bvalues = [0.005 0.05 0.10];

for k = 1:length(bvalues)

    b = bvalues(k);

    %DE
    f = @(t,y) a*y - b*y.^2 - (p*y.^3)./(y.^3 + q);

    %graph size constraints
    if b == 0.005
        tmax = 100;
        ymax = 160;
        ystep = 5;
        y0values = [10 50 100 140 155];

    elseif b == 0.05
        tmax = 40;
        ymax = 16;
        ystep = 0.5;

    % ICs
    y0values = [0 10 50 100 140 147.973 155];

    else
        tmax = 30;
        ymax = 8;
        ystep = 0.25;
        %%%%%why these numbers
        y0values = [0.25 0.75 1.2 3 6];
    end

    %make the grid
    tvals = linspace(0,tmax,25);
    yvals = 0:ystep:ymax;

    [T,Y] = meshgrid(tvals,yvals);

    %slope
    S = f(T,Y);

    %arrows
    U = ones(size(S));
    L = sqrt(U.^2 + S.^2);

    U = U./L;
```

```
    V = S./L;

    %make the figure
    figure(k)

    %direction feilds
    quiver(T,Y,U,V,0.5)
    hold on

    %plot the solutions
    for y0 = y0values

        [t,y] = ode45(f,[0 tmax],y0);

        plot(t,y,'LineWidth',2)

    end

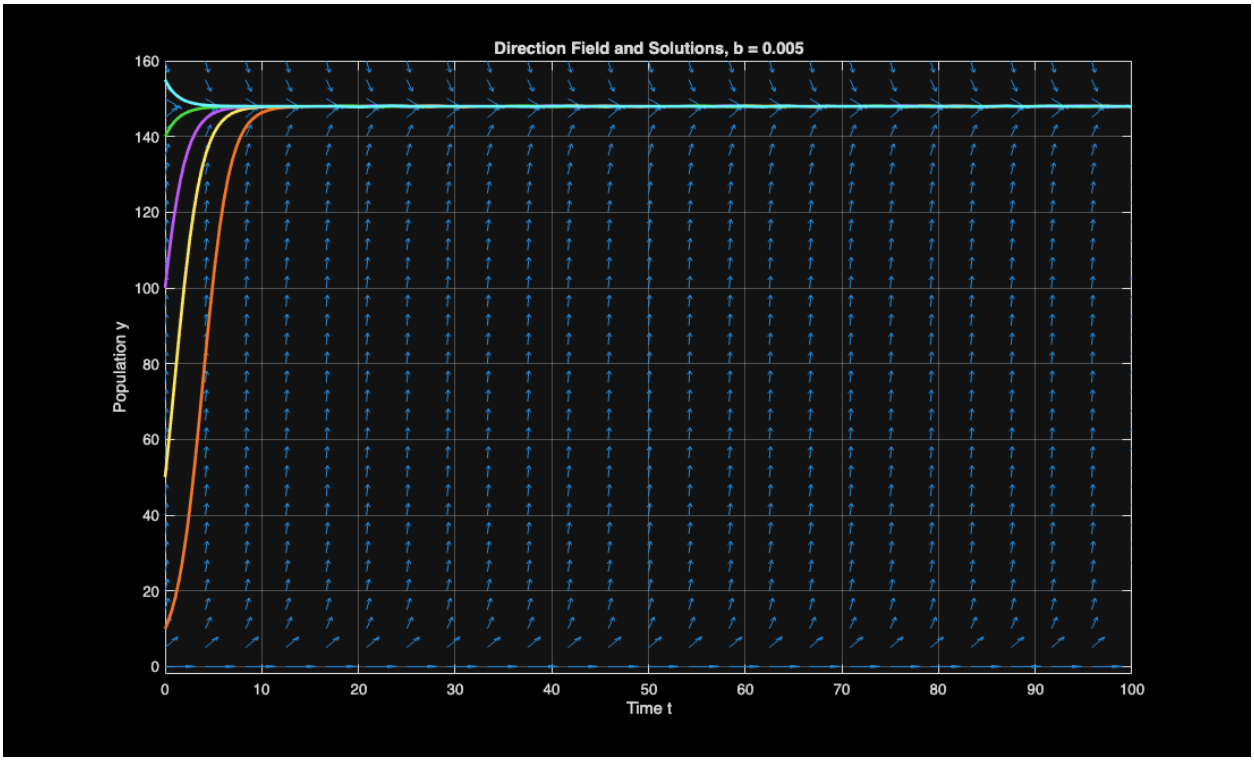
    %labels
    xlabel('Time t')
    ylabel('Population y')

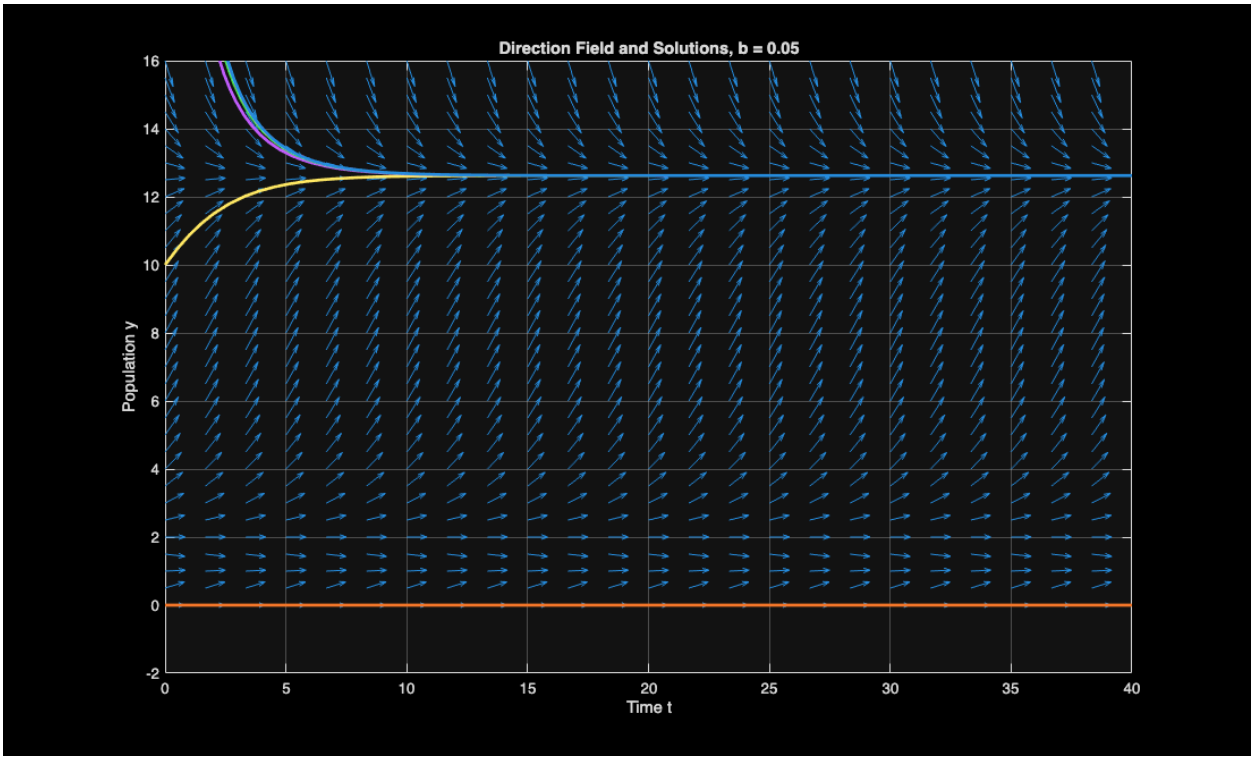
    %title
    title(['Direction Field and Solutions, b = ', num2str(b)])

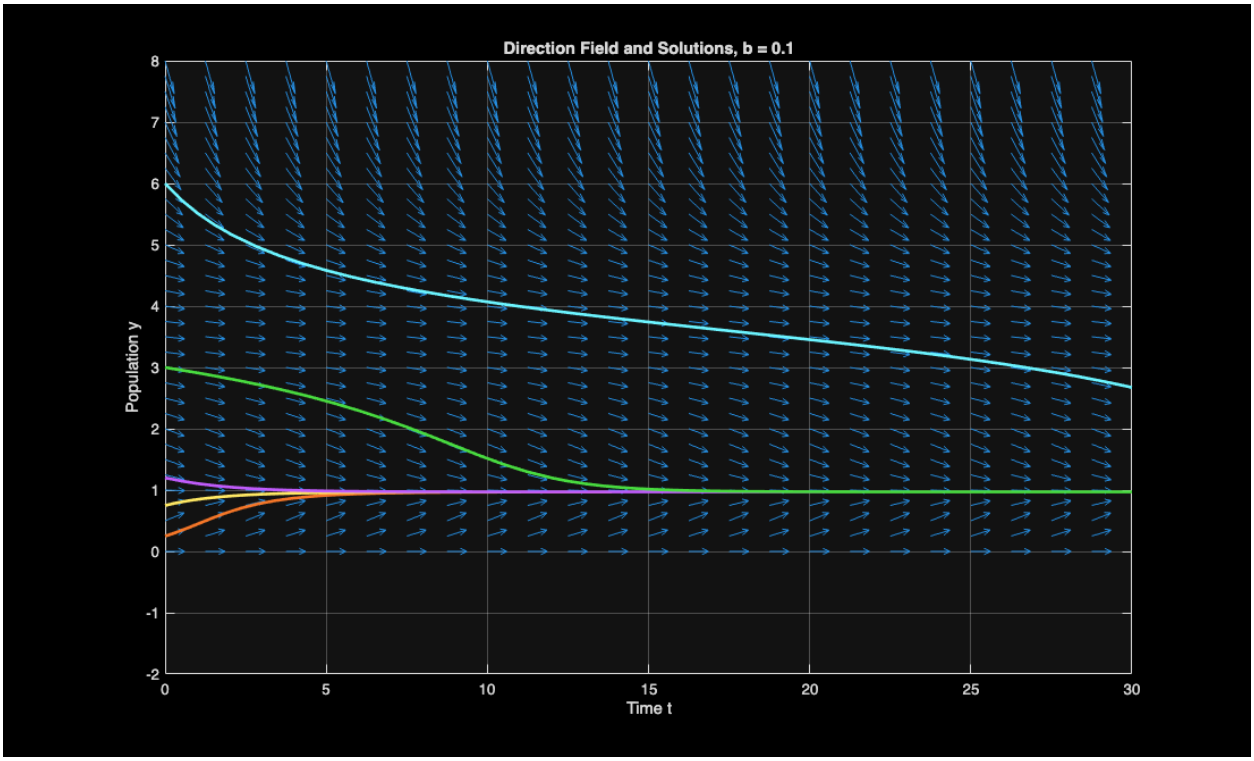
    %limits
    xlim([0 tmax])
    ylim([-2 ymax])

    grid on

end
```


Direction Field and Solutions, b = 0.005
Population y
Time t
0
20
40
60
80
100
120
140
160
0
10
20
30
40
50
60
70
80
90
100


Direction Field and Solutions, b = 0.05
Population y
Time t
-2
0
2
4
6
8
10
12
14
16
0
5
10
15
20
25
30
35
40



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